Schedule

Solutions to the midterm study questions are on LearnHub.

Case exam results will be announced next week.

Exams:

- Midterm: 19:00-21:00, Monday, May 5th
- No change in the topics covered

Labs:

- No lab this week
- Lab 6 will be next Thursday

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Agenda

Any questions on the previous material?

Hypothesis tests for categorical data

- Homogeneity
- Independence

Hypothesis tests for:

- Variances of two populations

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100 die rolls that you recorded

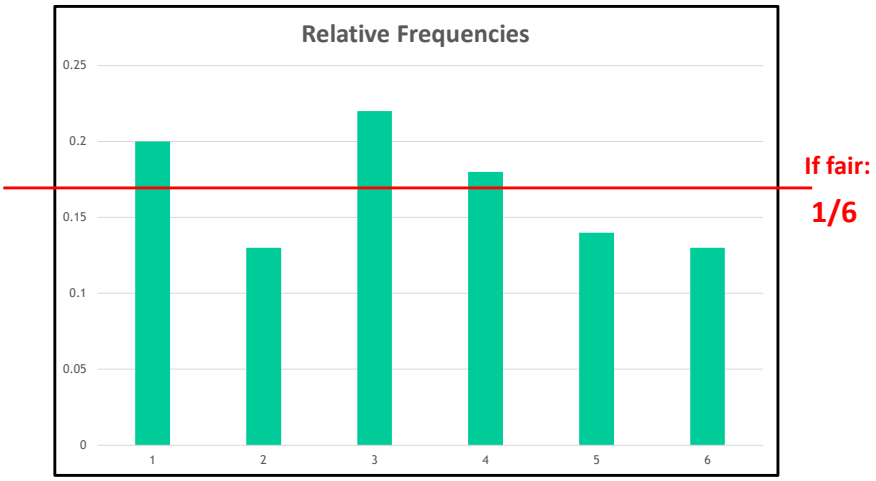
- How can you check whether the die is fair?

4	3	2	2	2	1	3	4	1	4
4	2	6	3	1	5	5	3	3	2
1	4	6	2	3	5	1	4	3	4
2	6	3	5	3	3	3	6	6	2
2	3	6	1	5	2	1	2	3	1
6	4	5	1	3	3	1	5	1	4
4	3	4	4	1	5	1	1	3	1
6	6	3	1	3	5	3	4	5	4
1	4	6	3	1	5	2	6	4	6
4	5	6	4	3	1	5	1	5	2

$$P(X = i) = \begin{cases} 1/6 & i = 1, \dots, 6 \\ 0 & otherwise \end{cases}$$

3

100 die rolls



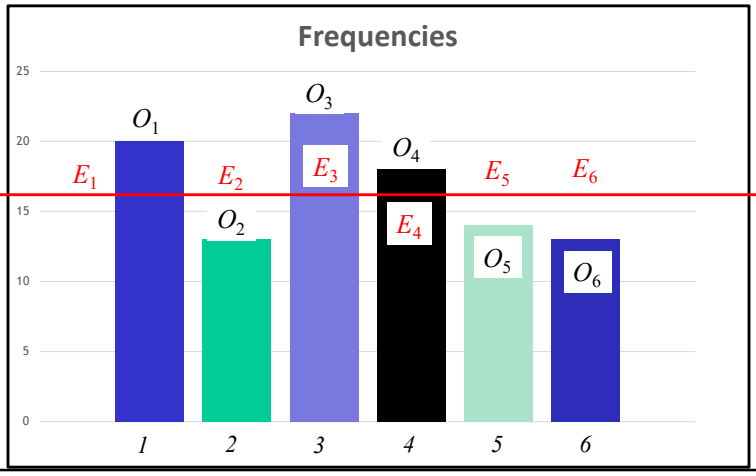
4

O_i: observed frequency in the *i*th class interval

E_i: expected frequency in the *i*th class interval if H_0 is true

E_i=16.67

100 die rolls



If fair:

16.67

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Goodness-of-fit test for a discrete uniform distribution

- H_0 : p1=p2=…=p6=1/6

  H_a : at least one pi ≠ 1/6

- n=100 observations in a sample
- k=6 number of outcomes

Assumption: $E_i \geq 5$ for all i=1,…,k.

observed frequency in the *i*th class interval

Test statistics:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{df}, \qquad df \text{ to be determined}$$

expected frequency in the *i*th class interval if H_0 is true

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Goodness-of-fit test for a discrete uniform distribution

- H_0 : p1=p2=…=p6=1/6

  H_a : at least one pi ≠ 1/6

Test statistics:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{k-1}$$

- n observations in a sample
- k number of outcomes
- Assume $np_i \geq 5$ ($E_i \geq 5$) for all i=1,…,k.

χ^2=4.52

$\chi^2_{6-1,0.1}=\chi^2_{5,0.1}$=9.24

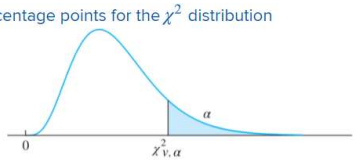
Rejection region:



R.R. : $\chi^2 \geq \chi^2_{k-1,\alpha}$

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TABLE A.7 Upper percentage points for the χ^2 distribution



						α				
ν	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156

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Test for Homogeneity

- Sometimes several multinomial trials are conducted, each with the same set of possible outcomes.
- The null hypothesis is that the probabilities of the outcomes are the same for each experiment.

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Contingency Tables

- Four machines manufacture cylindrical steel pins. The pins are subject to a diameter specification. A pin may meet the specification, or it may be too thin or too thick. Pins are sampled from each machine, and the number of pins in each category is counted. The following table contains the results:

	Too Thin	OK	Too Thick	Marginal total
Machine 1	10	102	8	120
Machine 2	34	161	5	200
Machine 3	12	79	9	100
Machine 4	10	60	10	80
Marginal total	**66**	**402**	**32**	**500**

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Contingency Tables

- The previous table is an example of a **contingency table**. Each row specifies a category regarding one criterion (machine, in this case), and each column specifies a category regarding another criterion (thickness, in this case). Each intersection of row and column is called a **cell**, so there are 12 cells in the table.

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Contingency Tables

- The number in the cell at the intersection of row *i* and column *j* is the number of trials whose outcome was observed to fall into row category *i* and into column category *j*. This number is called the **observed value** for cell *ij*.
- Note that we have included the totals of the observed values for each row and column. These are called the **marginal totals**.

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Test for Homogeneity

- The null hypothesis is that the proportion of pins that are too thin, OK, or too thick is the same for all machines.
- More generally, the null hypothesis says that no matter which row is chosen, the probabilities of the outcomes associated with the columns are the same.
- We will develop some notation with which to express H_0 and to define the test statistic.

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Test for Homogeneity

- Let I denote the number of rows in the table, and let J denote the number of columns. Let p_{ij} denote the probability that the outcome of a trial falls into column j given that it is in row i. Then the null hypothesis is

  H_0: For each column j, $\quad p_{1j} = \cdots = p_{Ij}$

- Let O_{ij} denote the observed value in cell ij. Let $O_{i.}$ denote the sum of the observed values in row i, let $O_{.j}$ denote the sum of the observed values in column j, and let $O_{..}$ denote the sum of the observed values in all the cells.

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Test for Homogeneity

- To define a test statistic, we must compute an expected value for each cell in the table. Under H_0, the probability that the outcome of a trial falls into column *j* is the same for each row *i*.
- The expected count for each cell is the proportion of trials whose outcome falls into column *j*, multiplied by the number $O_{i.}$ of trials in row *i*. That is

$$E_{ij} = \frac{O_{i.}O_{.j}}{O_{..}}$$

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Test for Homogeneity

- The test statistic is based on the differences between the observed and expected values:

$$\chi^2 = \sum_{i=1}^{I}\sum_{i=1}^{J}\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

- Under H_0, this test statistic has a chi-square distribution with $(I - 1)(J - 1)$ degrees of freedom.
- Use of the chi-square distribution is appropriate whenever the expected values are all greater than or equal to 5.

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Test for Homogeneity

- We can now test the null hypothesis that the proportions of pins that are too thin, OK, or too thick are the same for all the machines. The expected counts are:

	Too Thin	OK	Too Thick	Total
Machine 1	15.84	96.48	7.68	120.00
Machine 2	26.40	160.80	12.80	200.00
Machine 3	13.20	80.40	6.40	100.00
Machine 4	10.56	64.32	5.12	80.00
Total	**66.00**	**402.00**	**32.00**	**500.00**

Realized:

	Too Thin	OK	Too Thick	Total
Machine 1	10	102	8	120
Machine 2	34	161	5	200
Machine 3	12	79	9	100
Machine 4	10	60	10	80
Total	**66**	**402**	**32**	**500**

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Test for Homogeneity

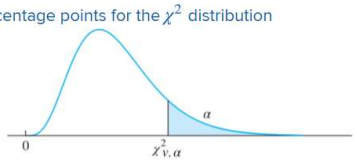
- All expected values are at least 5, so we compute the test statistic to be

$$\chi^2 = 15.5844.$$

- Under H_0, this test statistic has a chi-square distribution with (4 − 1)(3 − 1) = 6 degrees of freedom.

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TABLE A.7 Upper percentage points for the χ^2 distribution



	α									
ν	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
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9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156

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Test for Homogeneity

- All expected values are at least 5, so we compute the test statistic to be

$$\chi^2 = 15.5844.$$

- Under H_0, this test statistic has a chi-square distribution with (4 − 1)(3 − 1) = 6 degrees of freedom.
- To obtain the *P*-value, we consult the chi-square table (Table A.7). Looking under 6 degrees of freedom, we find that the upper 2.5% point is 14.449, and the upper 1% point is 16.812. Therefore 0. 01 < *P* < 0. 025.
- It is reasonable to conclude that the machines differ in the proportions of pins that are too thin, OK, or too thick.

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How to test independence?

Many times, the n elements of a sample from a population may be classified according to two different criteria. It is then of interest to know whether the two methods of classification are statistically independent;

An $r \times c$ Contingency Table

		Columns			
		1	2	…	c
Rows	1	O_{11}	O_{12}	…	O_{1c}
	2	O_{21}	O_{22}	…	O_{2c}
	⋮	⋮	⋮	⋮	⋮
	r	O_{r1}	O_{r2}	…	O_{rc}

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Example

Do different kinds of workers choose certain pension plans?
Are the two classifications independent?

Observed Data for Example

	Pension Plan			
Job Classification	1	2	3	Totals
Salaried workers	160	140	40	340
Hourly workers	40	60	60	160
Totals	200	200	100	500

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Contingency tables

- p_{ij}: probability that a randomly selected element falls in the *ij*th cell.
- u_i: probability that a randomly selected element falls in row class *i*. salaried vs hourly workers
- v_j: probability that a randomly selected element falls in column class *j*. pension plan, j=1,2,3
- If two classifications are independent: p_{ij} = ??

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Corresponding χ^2 test

H_0 : Two classifications are independent.
H_a : Two classifications are not independent.

- n: total number of observations
- Expected number of observations in cell (i,j): E_{ij} = ?

$\hat{u}_i$ =?
$\hat{v}_j$ =?

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij}-E_{ij})^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)}$$

- O_{ij} = ?
- RR = ?

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24

Example

Observed Data for Example

Job Classification	Pension Plan 1	2	3	Totals
Salaried workers	160	140	40	340
Hourly workers	40	60	60	160
Totals	200	200	100	500

Do different kinds of workers choose certain pension plans? Are the two classifications independent?

i	1	2
$\hat{u}_i$	340/500=0.68	160/500=0.32

i	1	2	3
$\hat{v}_j$	200/500=0.40	200/500=0.40	100/500=0.20

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Example

Table 9-3 Observed Data for Example 9-14

Job Classification	Pension Plan 1	2	3	Totals
Salaried workers	160	140	40	340
Hourly workers	40	60	60	160
Totals	200	200	100	500

Table 9-4 Expected Frequencies for Example 9-14

Job Classification	Pension Plan 1	2	3	Totals
Salaried workers	136	136	68	340
Hourly workers	64	64	32	160
Totals	200	200	100	500

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Contingency Table Tests

The eight-step hypothesis-testing procedure may now be applied to this problem.

1. The variable of interest is employee preference among pension plans.
2. H_0: Preference is independent of salaried versus hourly job classification.
3. H_1: Preference is not independent of salaried versus hourly job classification.
4. $\alpha = 0.05$
5. The test statistic is

$$\chi_0^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(o_{ij} - E_{ij})^2}{E_{ij}}$$

6. Since $r = 2$ and $c = 3$, the degrees of freedom for chi-square are $(r - 1)(c - 1) = (1)(2) = 2$, and we would reject H_0 if $\chi_0^2 > \chi_{0.05,2}^2 = 5.99$.

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Contingency Table Tests

7. Computations:

$$\chi_0^2 = \sum_{i=1}^{2} \sum_{j=1}^{3} \frac{(o_{ij} - E_{ij})^2}{E_{ij}}$$
$$= \frac{(160 - 136)^2}{136} + \frac{(140 - 136)^2}{136} + \frac{(40 - 68)^2}{68} + \frac{(40 - 64)^2}{64}$$
$$+ \frac{(60 - 64)^2}{64} + \frac{(60 - 32)^2}{32} = 49.63$$

8. Conclusions: Since $\chi_0^2 = 49.63 > \chi_{0.05,2}^2 = 5.99$, we reject the hypothesis of independence and conclude that the preference for pension plans is not independent of job classification. The P-value for $\chi_0^2 = 49.63$ is $P = 1.671 \times 10^{-11}$. (This value was computed using a hand-held calculator.) Further analysis would be necessary to explore the nature of the association between these factors. It might be helpful to examine the table of observed minus expected frequencies.

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Concluding remarks

- Goodness-of-fit tests: Important especially for simulation…
- How to test a certain probability distribution for a data set? Chi-squared test, Kolmogorov-Smirnov test…

  Statistics packages → Minitab, ARENA (more variety, e.g. Poisson)

  Simulation package
- How to select a probability distribution for a data set?

  Statistics packages → ARENA

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Inferences Concerning the true variance of a random variable X

- $X_1,\ldots,X_n$: random sample
- Assumptions:
  - $X_i \sim N(\mu, \sigma)$ ⟶ $\frac{(n-1)S^2}{\sigma^2} = \frac{\sum(X_i - \bar{X})^2}{\sigma^2} \sim \chi^2_{n-1}$

Normality is crucial, CLT does not apply here!!!

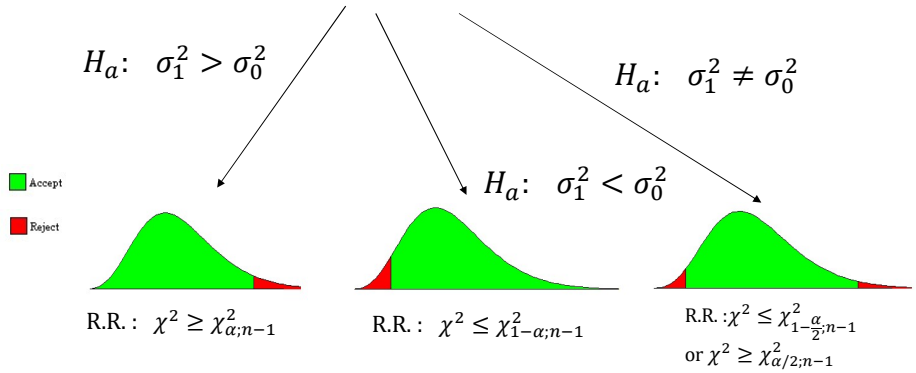
A new test statistics to check H_0: ${\sigma_1}^2 = {\sigma_0}^2$…

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Tests about a population Variance and Standard Deviation of a Normal r.v.

H_0: $\sigma_1^2 = \sigma_0^2$

Test statistics: $\chi^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$



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Inferences Concerning Two Population Variances

- Why do we need these tests?
- For some systems: A CHANGE IN THE VARIANCE OF PROCESS CAN BE IMPORTANT
  e.g., variance of contents in a bottle.
- Pooled t-test: CHECK ON THE EQUALITY OF VARIANCES

A new test statistics to check H_0: $\sigma_1^2 = \sigma_2^2$...

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Definition: F distribution

- Let *W* and *Y* be χ^2 random variables with *u* and *v* degrees of freedom, respectively, then:

$$\frac{W/u}{Y/v} \sim F_{u,v}$$

- Properties: $F_{u,v} = \frac{1}{F_{v,u}}$

$$F_{1-\alpha;u,v} = \frac{1}{F_{\alpha;v,u}}$$

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Example

- v_1=10, v_2=15,
- $F_{0.05,\ 10,15}$ = ?
- $F_{0.95,\ 15,10}$ = ?
- $P\{F_{10,15}>4.5\}$ = ?

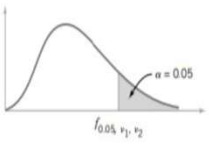


TABLE • VI Percentage Points f_{α,v_1,v_2} of the *F* Distribution (*Continued*)

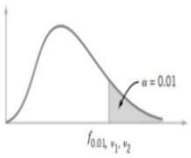
f_{α,v_1,v_2}

v_2 \ v_1	Degrees of freedom for the numerator (v_1)														
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.4
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.6
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.7
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.5
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.8
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.3
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.0
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.8
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.7
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.5
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.4
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.3
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.3
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.2
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.1
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.1
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.1
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.0

n for the denominator (v_2)

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Example



- $P\{F_{10,15}>3.8\} = ?$

TABLE • VI Percentage Points f_{α,ν_1,ν_2} of the *F* Distribution (*Continued*)

f_{α,ν_1,ν_2}

ν_2 \ ν_1	Degrees of freedom for the numerator (ν_1)														
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	
1	4052	4999.5	5403	5625	5764	5859	5928	5982	6022	6056	6106	6157	6209	6235	626
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.42	99.43	99.45	99.46	9
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.05	26.87	26.69	26.00	2
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.37	14.20	14.02	13.93	1
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.89	9.72	9.55	9.47	
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.31	
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	6.07	
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.28	
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.73	
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.33	
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	4.02	
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.78	
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.59	
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.43	
15	8.68	6.36	5.42	4.89	4.36	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.29	
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.18	
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.08	
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	3.00	
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.92	

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Theorem

$$\frac{W/u}{Y/v} \sim F_{u,v}$$

$X_1, \ldots, X_m \sim N(\mu_1, \sigma_1^2)$ and $Y_1, \ldots, Y_n \sim N(\mu_2, \sigma_2^2)$

$$\Rightarrow F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{m-1,n-1}$$

Proof: $$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum(X_i - \bar{X})^2}{\sigma^2} \sim \chi^2_{n-1}$$

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Corollary

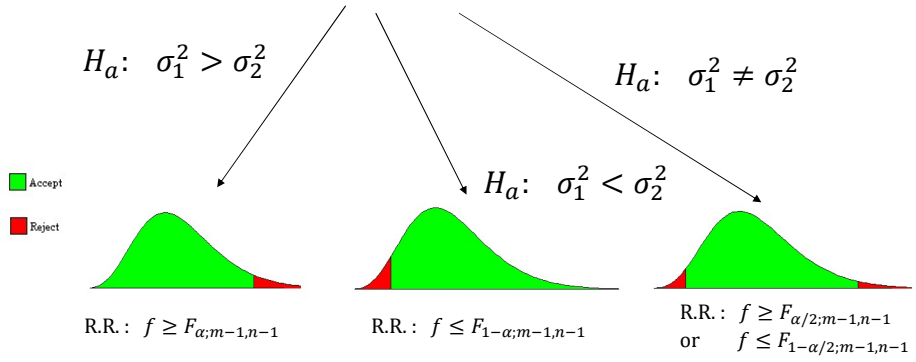
If $\sigma_1^2 = \sigma_2^2$, $\quad \dfrac{S_1^2}{S_2^2} \sim F_{m-1,n-1}$.

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Tests about the equality of two population variances (or standard deviations) of two Normal random variables

H_0: $\sigma_1^2 = \sigma_2^2$ $\qquad$ Test statistics: $F = \dfrac{S_1^2}{S_2^2}$



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